

Generation Interconnection System Impact Study Report

For Queue Project Q24-2380

Mecklenburg Power Station 345 kV
August 25, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2380. The Interconnection Customer proposes a 85.0 MW battery energy storage facility interconnecting at the Mecklenburg Power Station 345 kV substation in the Virginia area. The request was received on February 07, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 406 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$41,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- 1. Facility: Tiger Bay / Air Force Base 345.0 kV
Contingency: Artesia Country Club 345 kV/1 kV Autotransformer
Loading: 101.7% (Rating: 1,124 MVA)
DFAX: 7.5%
Pre-project loading: 101.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tiger Bay / Air Force Base 345.0 kV	Reconductor line	\$3,959,000	1% (\$41,000)
	Total allocated to project		\$41,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.